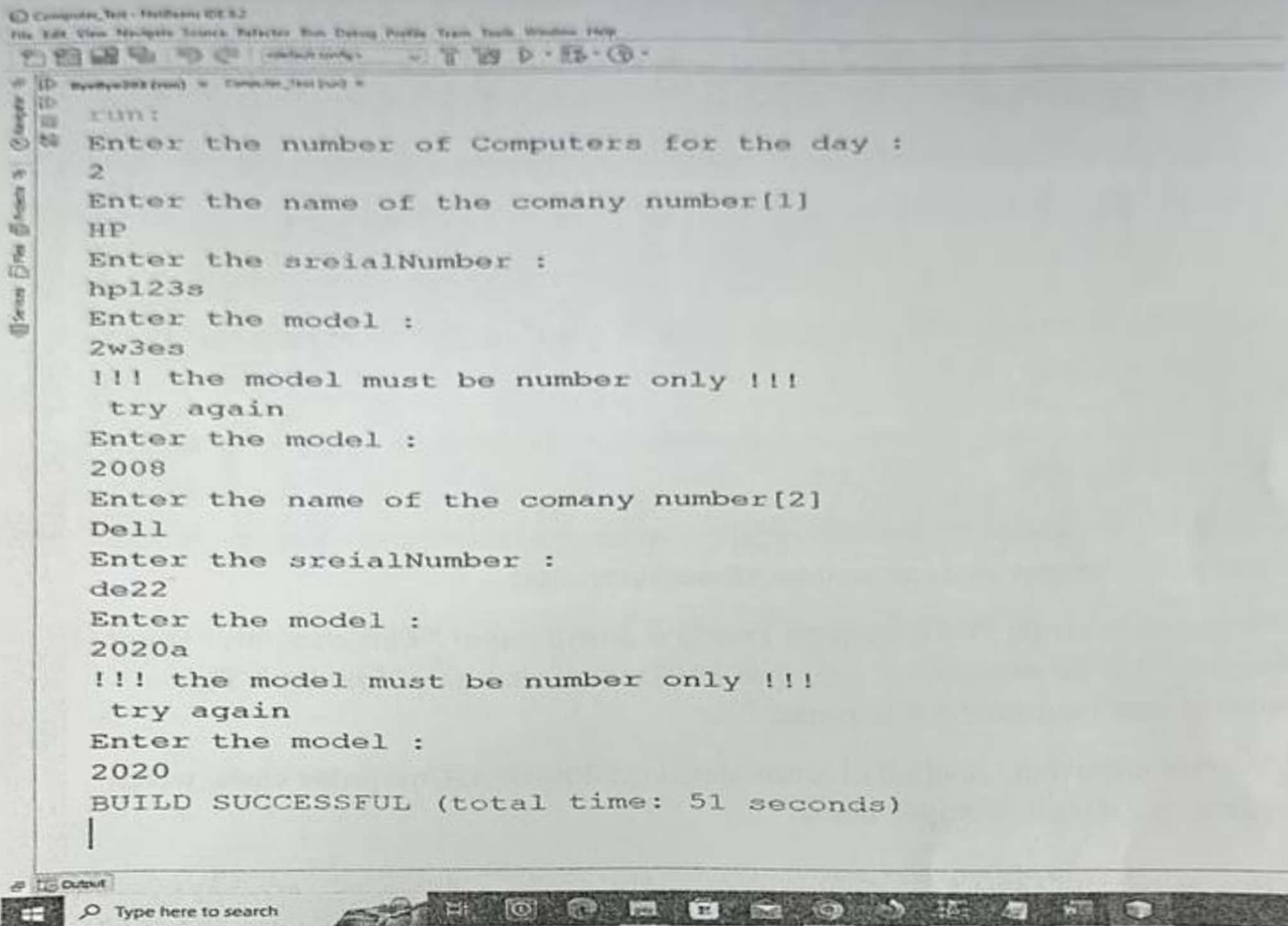


Final Lab Exam (07-12-2023)
Course : CPCS 203 (Programming II)

Student Name :

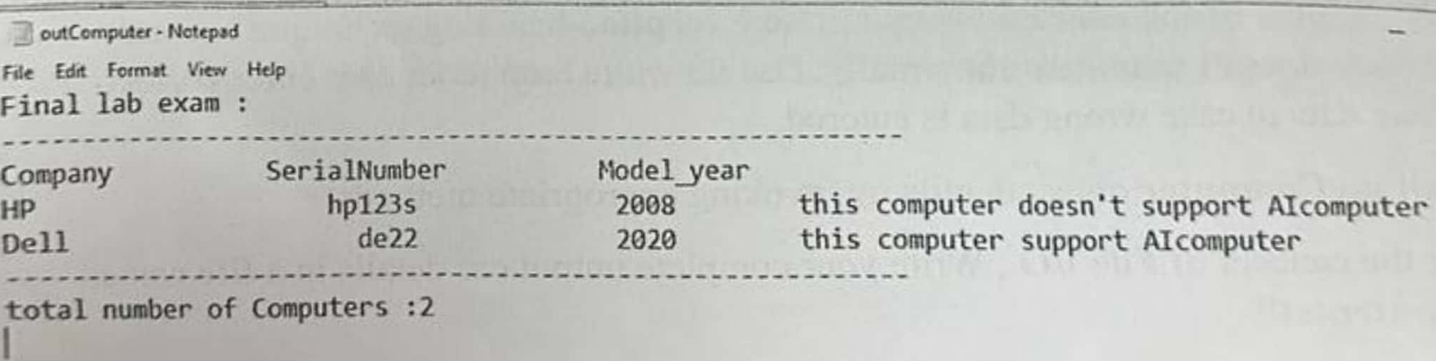
Student ID : _____ Section : _____

Sample out :



```
Computer_Test - IntelliJ IDEA 2023.2
File Edit View Navigation Sources Refactor Run Debug Profile Tools Windows Help
run
ID: MyJava2023 (run) x Computer_Test (run) x
run
Enter the number of Computers for the day :
2
Enter the name of the company number[1]
HP
Enter the serialNumber :
hp123s
Enter the model :
2w3es
!!! the model must be number only !!!
try again
Enter the model :
2008
Enter the name of the company number[2]
Dell
Enter the serialNumber :
de22
Enter the model :
2020a
!!! the model must be number only !!!
try again
Enter the model :
2020
BUILD SUCCESSFUL (total time: 51 seconds)
|
```

The file output :



```
outComputer - Notepad
File Edit Format View Help
Final lab exam :
-----
Company      SerialNumber      Model_year
HP            hp123s            2008            this computer doesn't support AIcomputer
Dell          de22              2020            this computer support AIcomputer
-----
total number of Computers :2
|
```

1. Create a **Computer** class, which is **abstract** super class of all cars.

- a) **Computer** class includes TWO attributes **String Company** and **serialNumber**.
- b) A **constructor** to initialize all the data members of **Computer** class.
- c) Override **toString()** method to return details about the **Computer** class.

2. Create an **interface Computer5g** with the following details:

- a) Declare a constant **SmartWiFi** of type **Boolean** with value **true**.
- b) Declare a method called **WiFiSupport()** which returns a **String** value .

3. Create the **AIcomputer** class that inherits **Computer** class and deals with **Computer5g** interface :

- a) **AIcomputer** class must include TWO attributes to store the **modelYear** of type **integer** and **count of** type **static int** .
- b) Define a **constructor** that takes appropriate parameters so as to initialize data members of **Computer** class as well as **AIcomputer** class .
- c) When you override **WiFiSupport()** method ,it will return "**This computer support AIcomputer** " if the **modelYear** over than **2015** otherwise it should return " **This computer doesn't support AIcomputer** " .
- d) Override **toString()** method to return details about the **AIComputer** class ,which also includes the details of **super class** .

4. Create **ComputerTester** class to test your application:

- a) Declare and instantiate a variable **myComputer** to be an empty **ArrayList** of type **AIcomputer** , [All the data must be entered by **the user**. The **size** of **ArrayList** depends on the first number of **the user entering** it , Use **Exception** handling technique to make sure your code doesn't terminate abnormally. Use **do-while** loop to let user enter correct **model_Year** data in case wrong data is entered.
- b) List all **myComputer** object details by invoking appropriate methods.
- c) Using the concept of **File I/O** , Write your complete output car details in a **file** named **outComputer.txt** .